

■ SOLUTION BRIEF

Why TriMagnetix

Nanomagnetic Logic. Extreme power efficiency, unmatched reliability, drop-in integration with existing chip infrastructure.

800× lower energy than comparable CMOS bits.
1000 W ON CMOS → 1.25 W ON NML

01

Extreme Power and Thermal Efficiency

-  **Proven 800× Lower Power:** We've documented an 800× power reduction using NML vs comparable CMOS bits. Leveraging new circuit-design paradigms enabled by NML, we aim to lower system power consumption even further.
-  **Zero Standby Power:** NML is inherently non-volatile, effectively eliminating standby power leakage. It also allows for new architectures like in-memory compute, reducing energy and time needed for data transfer between memory and logic.
-  **Ultra-Low Heat:** Our system generates orders of magnitude lower heat than traditional silicon, which is vital for deployment in compact, isolated environments where cooling is difficult.

02

Unmatched Reliability and Performance

-  **Radiation Resistance:** NML devices offer inherent radiation resistance, a crucial feature for aerospace hardware.
-  **Error Mitigation:** We are overcoming traditional noise challenges by using a novel hybrid architecture. This architecture integrates our patented NML triangle with Magnetic Tunnel Junctions (MTJs) and supports active, clocked signal restoration between logical blocks.
-  **High Speed Potential:** Our technology is compatible with and will utilize Spin-Transfer Torque (SOT), Voltage-Controlled Magnetic Anisotropy (VCMA), and beyond, demonstrating high-speed switching in the picosecond regime.

03

Leverages Existing Infrastructure and Tooling

-  **Drop-in Replacement:** TriMagnetix's NML technology serves as a drop-in replacement for existing processor technology, allowing it to be integrated as an ASIC into larger chips used in the field.
-  **Fast Development and Iteration:** By leveraging "Minimal Fab" methods, we can provide Proof-of-Concept (PoC) builds and bespoke chips in just 2–3 weeks, significantly faster than the multiple months required by traditional manufacturers.

■ BY THE NUMBERS

800×

LOWER POWER
VS CMOS

2–3^{wk}

POC TURNAROUND
BESPOKE CHIPS

0^w

STANDBY
POWER LEAKAGE

ps

SWITCHING SPEED
REGIME

■ THE TECHNOLOGY

-  **Patented triangle NML:** logic and memory merged into one magnetic element, switched by electrical pulses instead of constant current.
-  **Hybrid architecture:** NML triangles paired with Magnetic Tunnel Junctions (MTJs) for clocked signal restoration and noise mitigation.
-  **Picosecond switching:** compatible with Spin-Transfer Torque (STT) and Voltage-Controlled Magnetic Anisotropy (VCMA) for high-speed operation.
-  **Drop-in ASIC:** integrates into existing chip stacks without redesigning the surrounding process flow.



SCAN TO CONNECT

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